

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-9223
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture: Automated Standard Laptop Procurement using ServiceNow Flow Designer

Reference: <https://www.servicenow.com/docs>

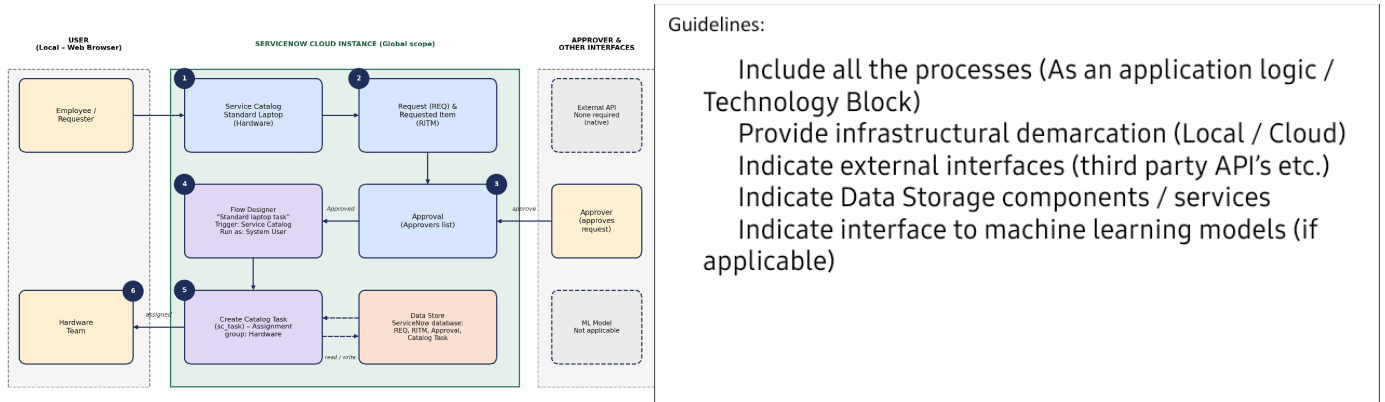


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employee / Requester orders the laptop from the ServiceNow Service Catalog (Service Catalog → Hardware → Standard Laptop → Order Now) and tracks the Request number; Approver and Hardware team use the ServiceNow web UI.	ServiceNow Web UI (HTML, CSS, JavaScript rendered by the platform), Service Catalog
2.	Application Logic-1	Standard Laptop catalog item (Lenovo Carbon x1) creates a Request (REQ) and Requested Item (RITM); Process Engine of the item is set to the "Standard laptop task" flow.	ServiceNow Service Catalog, Maintain Items
3.	Application Logic-2	Approval of the request: the approver approves the request from the Approvers section so that the Requested Item holds the approval record.	ServiceNow Approvals
4.	Application Logic-3	Automation: flow "Standard laptop task" (Global, Run as System User, trigger: Service Catalog) uses the Create Catalog Task action to create the task for the Requested Item with Short Description and Description "Laptop need to Configured", Approval "Approved" and Assignment group "Hardware".	ServiceNow Flow Designer, Create Catalog Task action (ServiceNow Core)
5.	Database	Stores the Request, Requested Item, Approval and Catalog Task [sc_task] records, including Short description, Description, Assignment group and Approval fields.	ServiceNow platform database (tables: sc_request, sc_req_item, sysapproval_approver, sc_task)
6.	Cloud Database	Database service provided with the ServiceNow cloud instance; no separate database is set up for this project.	ServiceNow cloud-hosted instance database
7.	File Storage	No separate file storage is required; any attachments added to a request or task are stored with the record.	ServiceNow platform attachments
8.	External API-1	Not required – the complete workflow is configured inside ServiceNow without a third-party API.	None (native ServiceNow)
9.	External API-2	Not required.	None

10.	Machine Learning Model	Not applicable – the automation is rule-based (trigger and action), without a machine learning model.	Not applicable
11.	Infrastructure (Server / Cloud)	Application Deployment on Cloud: ServiceNow cloud instance (Global application scope), accessed through a web browser. Local Server Configuration: Not required – users only need a computer with an internet connection and a web browser. Cloud Server Configuration: ServiceNow-hosted instance with Flow Designer, Service Catalog and Approvals available.	ServiceNow Cloud (Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is developed for this project; the solution is configured using the low-code tools of the ServiceNow platform.	ServiceNow Flow Designer (low-code), Service Catalog
2.	Security Implementations	Access to the catalog, approvals and tasks is controlled through ServiceNow login and role-based access control. Task creation depends on the approval process, and the flow runs as System User in Global scope.	ServiceNow roles and ACLs, approval process, HTTPS
3.	Scalable Architecture	The flow creates a Catalog Task for every Requested Item automatically, so more requests do not add manual work. The platform separates the user interface, application logic (Flow Designer) and database, and the flow can be reused for other catalog items.	ServiceNow multi-tier platform, Flow Designer
4.	Availability	The application is hosted on the ServiceNow cloud platform and is available through a web browser to requesters, approvers and the Hardware team without local installation.	ServiceNow cloud-hosted instance

S.No	Characteristics	Description	Technology
5.	Performance	The flow is event-driven: it starts when the trigger condition is met and creates the Catalog Task without manual steps. Flow execution can be monitored in the Flow Designer Executions tab.	Flow Designer trigger and action, Executions log

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>